

Ensemble Methods: Boosting vs. Bagging

An Empirical Study Across Four Datasets

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AI Academy – Machine Learning Final Project, Spring 2026

July 2026

Central Question

*Under what conditions does boosting outperform bagging, and vice versa — and **why**?*

Boosting (AdaBoost / SAMME)

- Re-weights hard examples sequentially
- Drives down bias
- Sensitive to label noise

Bagging (Random Forest)

- Bootstrap aggregation, trained independently
- Drives down variance
- More robust to noise

- All algorithms implemented from scratch in NumPy (no `sklearn` ensemble classes)
- Verified against `sklearn` on every dataset; 190 tests, 95% line coverage

Decision Tree (CART)

- Gini impurity or entropy splitting
- Weighted samples (for AdaBoost);
max_features subsampling (for RF)
- Matches sklearn within 2% on all four datasets

AdaBoost (SAMME)

- Depth-1 stumps, re-weighted each round:
$$\alpha_m = \ln \frac{1-\epsilon_m}{\epsilon_m} + \ln(K-1)$$
- Misclassified samples get more weight next round
- Bonus: one-vs-rest wrapper, SAMME.R

Random Forest

- T trees on bootstrap samples; each split considers $\lfloor \sqrt{p} \rfloor$ random features
- Majority vote; out-of-bag (OOB) scoring as a free validation proxy
- Parallel training via `multiprocessing.Pool`

PCA / K-Means / DBSCAN

- PCA: eigendecomposition of covariance, scree + 2D projection
- K-Means: Lloyd's algorithm, `k-means++` init, elbow method
- DBSCAN: density-based; ϵ from k -distance knee

Datasets & Experimental Design

Dataset	Samples	Features	Task
Breast Cancer Wisconsin	569	30	Binary (63/37)
Credit Card Fraud	284,807	30	Binary, 0.17% fraud
MNIST (3 vs. 8 subset)	6,000	784	Binary, high-dim
Covertypes (subset)	15,000	54	7-class

- Satisfies brief: severe imbalance ($\leq 1\%$) + high-dimensional (> 20 features)
- Features standardised; SMOTE applied to Credit Card Fraud training fold only
- **7 experiments**, seeded (`random_state=42`), reproducible via python `src/experiments/run_all.py`

Exp 1 – Baseline Comparison

Unpruned tree, depth-1 stump, and sklearn's tree on identical 80/20 splits.

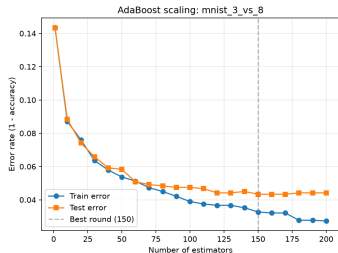
Dataset	Ours	sklearn	Diff.
Breast Cancer	0.9123	0.9123	0.0000
Credit Card Fraud	0.9991	0.9991	0.0000
MNIST (3 vs. 8)	0.9558	0.9583	0.0025
Covertypes	0.7367	0.7370	0.0003

Max. gap 0.25 points (MNIST) – well within the 2% tolerance. A single stump already reaches 0.921 on Breast Cancer but only 0.639 on Covertypes, motivating ensembling.

Exp 2 & 3 – Scaling: AdaBoost & Random Forest

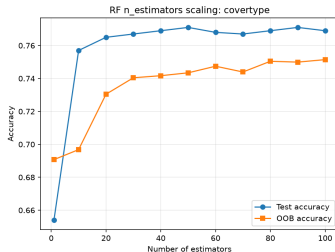
AdaBoost ($n_estimators : 1 \rightarrow 200$)

- Training error $\rightarrow 0$ within 30–40 rounds
- Test error plateaus early (best round 10–150)
- No overfitting up to 200 rounds



Random Forest ($n_estimators : 1 \rightarrow 100$)

- Accuracy rises sharply to ~ 20 trees, then flat
- OOB accuracy tracks test accuracy within 1–2 points
- Deeper trees help most on Covertypes / MNIST



Exp 4 – Head-to-Head Comparison

5-fold CV, $n_estimators = 100$, mean $\pm$ std accuracy:

Dataset	Tree	AdaBoost	Our RF	sklearn RF
Breast Cancer	.910 $\pm$.019	.967$\pm$.018	.951 $\pm$.014	.956 $\pm$.012
Credit Card	.992 $\pm$.001	.995 $\pm$.001	.998$\pm$.001	.998$\pm$.001
MNIST (3v8)	.952 $\pm$.003	.950 $\pm$.004	.979 $\pm$.003	.980$\pm$.002
Covertypes	.687 $\pm$.010	.623 $\pm$.014	.778 $\pm$.014	.784$\pm$.006

AdaBoost wins on clean, balanced Breast Cancer; Random Forest wins (or matches sklearn) on all three harder datasets – widest gap on Covertypes (15 points).

Exp 5 & 6 – Noise Robustness & Bias-Variance

Noise Robustness (flip up to 20% of labels)

- Breast Cancer, MNIST: AdaBoost degrades 3–6× faster than RF
- Credit Card, Covertypes: pattern reverses, but gap <1.6 points

Bias–Variance ($B=100$, Breast Cancer)

Model	Bias ²	Variance
Single Tree	0.0702	0.0626
AdaBoost	0.0351	0.0257
Random Forest	0.0585	0.0146

AdaBoost halves bias²; Random Forest quarters variance – the textbook mechanism.

Exp 7 & Bonus – Unsupervised Analysis, SAMME.R vs. OvR

Unsupervised (PCA / K-Means / DBSCAN)

- Breast Cancer: K-Means ARI 0.654 – classes well separated
- Credit Card Fraud: K-Means ARI ≈ 0 – fraud invisible to distance clustering
- Covertypes: DBSCAN (0.160) beats K-Means (0.044) – non-convex geometry

SAMME.R vs. OvR (Covertypes, ~ 100 stumps)

- SAMME.R: 0.674 acc., log-loss 2.650, 4.62s
- OvR (compute-matched): 0.678 acc., 3.93s – essentially tied
- Naive OvR ($7\times$ stumps): +2 acc. points for $7\times$ the time – poor trade-off

Key Findings & Conclusion

- **Boosting wins:** binary, reasonably separable classes, clean labels – lowest bias², best accuracy on Breast Cancer
- **Bagging wins:** severe class imbalance, high dimensionality, or >2 classes – lowest variance, native multi-class handling
- Implemented Decision Tree, AdaBoost (SAMME), Random Forest, PCA, K-Means, DBSCAN from scratch (NumPy only); verified vs. sklearn within 2%; 190 tests, 95% line coverage
- **Limitations:** RF parallelism untested at scale; noise-robustness reversal on 2 datasets needs deeper study
- **Future work:** from-scratch Gradient Boosting Machine; extend bias-variance decomposition to all four datasets

Thank you – Questions?